SMART INDIA HACKATHON 2026

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- **Problem Statement ID** - 26218
- **Problem Statement Title** - [Student Innovation] Designing Drones and Robots to solve Pressing Challenges in India .....
- **Theme** - Robotics and Drone
- **PS Category** - Hardware
- **Team ID** -
- **Team Name** - Coders Creed

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BREAKING NEWS

INDIA'S WATER BODIES ARE UNDER PRESSURE

Pollution is not just contaminating our water -
it is making monitoring and cleanup increasingly difficult

CLEAN WATER HEALTHIER INDIA

PROBLEM STATEMENT

Develop intelligent Drones and Robotic systems to solve real-world problems through autonomous innovation.

Address the pressing challenges faced by India with scalable, affordable and impactful technology



THE SCALE

72,368 MLD
Urban wastewater generated per day

28%
Actually treated

72%
Remains Untreated

Source : NITI Aayog | Urban Wastewater Scenario in India (CPCB , 2020-21)

WHY IT HAPPENS

- Untreated / inadequately treated sewage
- Solid waste & dumping
- Rapid urbanisation
- Inadequate sewerage networks
- Poor operation & maintenance
- Monitoring gaps



Pollution >>>
Detection>>>
Response gap

The CONSEQUENCES

WATER QUALITY
Contamination of water bodies

ECOSYSTEM
Damage to aquatic ecosystems

HUMAN EXPOSURE
Unsafe manual cleaning & health risks

MONITORING GAP
Pollution hotspots hard to detect & track

THE TECHNOLOGY GAP

- Continous Monitoring
- Intelligent Detection
- Targeted Response
- Autonomous action

The Gap is not only Treatment -
It is Intelligent , Continous Detection and Targeted Action

This is where our solution enters.

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NEERKAVALAN

INTELLIGENT GUARDIAN OF WATER BODIES

CLEAN WATER HEALTHIER INDIA

PROPOSED SOLUTION

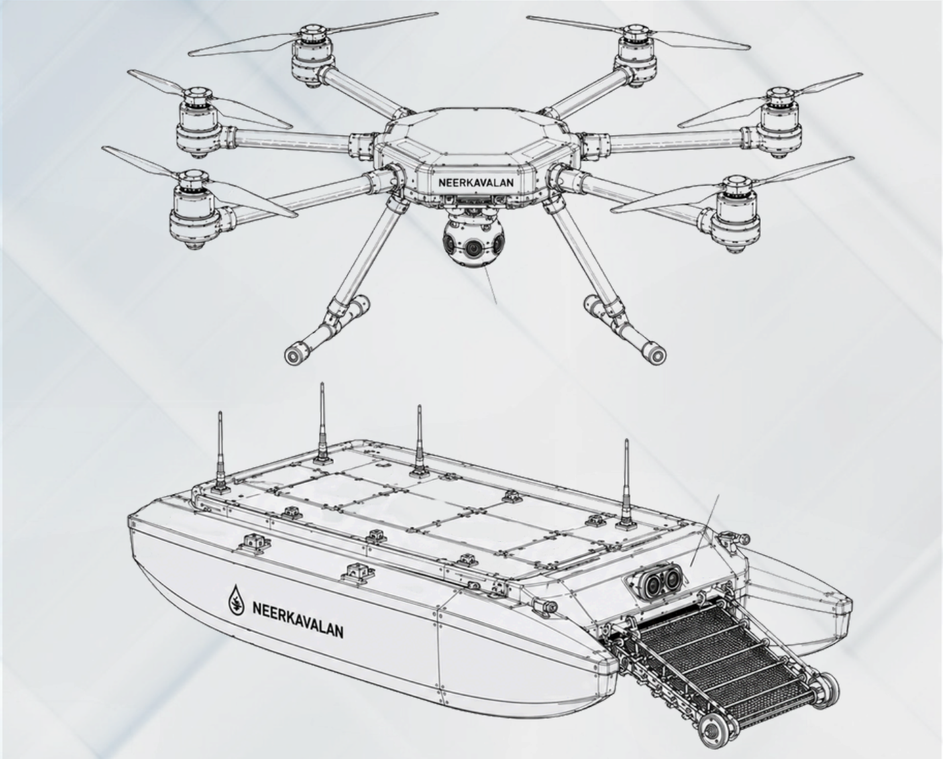
Neerkavalan is an intelligent **Drone and USV** ecosystem that transforms water-body cleaning from manual & reactive to autonomous & targeted.

HOW IT ADDRESSES THE PROBLEM

- Aerial Surveillance
- Autonomous Navigation
- Surface Waste Collection
- Real Time Monitoring
- AI-Driven Intelligence

DETECT → MAP → NAVIGATE → COLLECT → MONITOR

END - TO - END INTELLIGENT WATER BODY MANAGEMENT



NEERKAVALAN MODEL

UNIQUENESS

AIR — OBSERVE → DETECT — ANALYZE → WATER — ACT → ACTION — CLEAN

CURRENT APPROACH	OUR ADVANTAGES
Manual Monitoring	Continous Monitoring
Manual waste collection	Targeted Cleaning
Limited coverage	Wide Area Coverage
Delayed Detection	Early Detection
High Human Effort	Autonomous Operation

INNOVATION

Dual Robot System
Drone + USV working

Intelligent Detection
AI Model Detection & Classification

Targeted Response
USV Deployed where only needed

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SYSTEM ARCHITECTURE

1. MISSION PLANNING

1. Define Lake Boundary
2. Divide Lake into Sectors
   A | B | C | D
   A B C D
3. Generate GPS Waypoints for Each Sector
4. Upload Waypoints to Drone (Pixhawk)
5. Start Mission

6. DRONE SYSTEM (UAV)

Onboard Sensors
- DJI Air2S Camera
- GPS (M10)
- Battery (11.1V 5200mAh)

Raspberry Pi 5 (On Drone)
- Image Capture (Real-time)
- YOLO26n Detection (8 Classes)
- Waste / Plant Detected?
- Geolocation (Localization)
- MQTT Client (PUBLISH)

Pixhawk 6C Mini (Flight Controller)
- Autonomous Flight
- Follow Waypoints
- RTL / Land / Recharge Logic

Drone Internal Communication
Raspberry Pi 5 ↔ MAVLink UART ↔ Pixhawk 6C Mini ↔ Motors (EMAX MT2213)

7. COMMUNICATION

- MQTT Client (PUBLISH)
- 4G / LTE (Internet)
- MQTT BROKER (Cloud / Server)
- Laptop SUBSCRIBER
- Boat Raspberry Pi SUBSCRIBER

8. BOAT SYSTEM (USV)

Raspberry Pi 5 (On Boat)
- MQTT Client (SUBSCRIBE)

- Receive Location (Target)
- GPS Navigation (PM02)
- Motor Controller /Pixhawk
- Boat Motors (A2212)
- Move to Location
- Local Search (Boat Camera + AI)
- Waste Found? (No → Search More; Yes)
- Collection Mechanism (Conveyor / Cutter)
- Collect Waste / Remove Plants
- Update Status (PUBLISH)

9. LAPTOP DASHBOARD

MQTT Client (SUBSCRIBE)

- Live Drone Location
- Live Boat Location
- Sector Status (A/B/C/D)
- Waste Location on Map
- Waste Type & Confidence
- Drone / Boat Battery
- RF Link Status
- 4G / Internet Status
- Live Camera Feeds
- Mission Progress
- Alerts & Notifications

A B C D

Drone Camera Feed (DJI Air2S)

Boat Camera Feed

TECH IMPLEMENTATION (STACK)

Hardware
- Raspberry Pi 5
- Pixhawk 6C Mini
- M10 GPS
- SiK Telemetry
- 4G / LTE Modem
- Cameras
- Motors & ESC

Flight Stack
- PX4 Autopilot
- MAVLink

AI Model
- YOLO26n
- Ultralytics 8.4.133
- 8 Classes
- 640x640 Input

Communication
- MQTT
- SiK Radio (Direct Link)
- 4G / LTE (Cloud Link)

Backend
- FastAPI (Python)
- PostgreSQL + PostGIS
- MQTT Broker (EMQX)
- WebSocket
- REST API

Frontend
- Next.js (React)
- TypeScript
- Leaflet / Mapbox
- Tailwind CSS
- Charts

DevOps
- Docker / Docker Compose
- GitHub Actions (CI/CD)
- Systemd (Edge)
- Nginx (Reverse Proxy)

10. OPERATION SEQUENCE (STEP-BY-STEP)

1. Drone takes off and goes to Sector A autonomously
2. Scans Sector A using Camera + YOLO
3. Waste / Plants detected
4. Get GPS location
5. Publish to MQTT Broker
6. Boat receives location and navigates to A
7. Boat reaches and collects waste/plants
8. Sector A Completed
9. Drone moves to Sector B
10. Repeat for Sectors B, C and D (A B / C D)

Repeat Until All Sectors Are Completed

11. MISSION COMPLETE

All Sectors Cleaned

Final Report & Data Analysis

TECHSTACK

Python · Google colab · YOLO · TensorFlow · Google Cloud · node.js · ArduPilot

FRONTEND

live location

live video streaming

Manual operation is available in case of emergency

Our prototype is now 58% completed with core features . We're entering the testing and validation phase with integration